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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

IS3241-1 – CLOUD COMPUTING

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The task level parallelism is parallelized by
 - A) Programmer
 - B) Compiler
 - C) Processor
 - D) Operating System
2. _____ describes a cloud service that can only be accessed by a limited amount of people.
 - A) Public Cloud
 - B) Private Cloud
 - C) Hybrid Cloud
 - D) Data Centre
3. _____ computing refers to applications and services that run on a distributed network using virtualized resources.
 - A) sequential
 - B) Mainframe
 - C) cloud
 - D) Parallel
4. The _____ cloud infrastructure is operated for the exclusive use of organization.
 - A) Public
 - B) Private
 - C) Hybrid
 - D) Data center
5. The capability provided to the consumer to use provider's applications running on cloud infrastructure is _____.
 - A) XaaS
 - B) IaaS
 - C) PaaS
 - D) SaaS
6. _____ defining a service provisioning model for compute services in which resources are packaged and offered on a pay-per-use basis.
 - A) Utility-Oriented computing
 - B) Grid Computing
 - C) Cluster computing
 - D) Mainframes
7. VMM facilitates sharing of _____.
 - A) memory and I/O
 - B) CPU and memory
 - C) CPU, memory and I/O
 - D) CPU and I/O
8. In a _____ scheme, the VM is installed as type 1 hypervisor directly onto the hardware.
 - A) Full Virtualization
 - B) Para Virtualization
 - C) Partial Virtualization
 - D) Emulator
9. Which of the following virtualization types is primarily used to deliver a virtualized desktop environment to end-users?
 - A) Network virtualization
 - B) Desktop virtualization
 - C) Storage virtualization
 - D) Application virtualization
10. _____ is a system administration practice that allows decoupling the physical organization of the hardware from its logical representation.
 - A) Storage Virtualization
 - B) Desktop Virtualization
 - C) Network Virtualization
 - D) Application-Server Virtualization

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11. What is the role of a virtual machine monitor (VMM)?
 A) To manage virtual machines and mediate access to hardware resources
 B) To store data securely in a virtual environment
 C) To manage cloud service pricing
 D) To provide network-level encryption
12. Azure Storage Account provides which of the following storage types?
 A) Blob, Azure Drive, Queue, Table
 B) SQL, Blob, Azure Drive, Queue
 C) Cosmos DB, Blob, Azure Drive, Table
 D) Blob, Azure Drive, Queue, Redis
13. In Google App Engine, which service allows you to store and retrieve large amounts of structured data?
 A) Datastore
 B) Blobstore
 C) Cloud Storage
 D) BigQuery
14. What is the maximum size of an individual S3 object?
 A) 5 GB
 B) 10 TB
 C) 5 TB
 D) 10 GB
15. In a market-driven cloud, the concept of elasticity refers to:
 A) The flexibility of data storage
 B) The ability to scale resources up or down
 C) The physical expansion of data centers
 D) The adjustment of network bandwidth
16. One of the challenges of Fog Computing is:
 A) Lack of scalability
 B) Complexity in managing distributed resources
 C) High centralization of resources
 D) Reduced data privacy
17. In a microservices architecture, services typically communicate with each other using:
 A) Synchronous communication only
 B) Static IP addresses
 C) HTTP/REST or messaging queues
 D) Physical network cables
18. What is Network Function Virtualization (NFV) primarily used for?
 A) Increasing the physical size of data centers
 B) Virtualizing desktop applications
 C) Reducing network security
 D) Decoupling network functions from dedicated hardware devices
19. Which component in an SDN architecture is responsible for managing network policies?
 A) SDN controller
 B) Network switches
 C) Data plane
 D) End-user devices
20. Micro services is most closely related to what other established acronym?
 A) SOA
 B) API
 C) ABI
 D) ISA

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the key features of cloud computing according to NIST's definition. Explain the ways in which these traits help organizations.	8	L*2	1	1,2
	b) Examine how computing has changed over time, from mainframes to cloud computing. Describe the key turning points that caused cloud services to arise.	8	L2	1	1,2
2.	a) Describe the key principles of parallel and distributed computing. How do these principles improve scalability and performance?	8	L2	2	1,2
	b) Explain Remote Procedure Call reference model with neat diagram.	8	L2	2	1,2
3.	a) With a neat diagram explain the hypervisor reference architecture.	8	L2	2	1,2
	b) Define virtualization. Evaluate the pros and cons of virtualization in the context of cloud computing.	8	L3	2	1,2

Unit – II

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|-------|--|---|----|---|-----|
| 4. a) | Discuss the cloud reference model. How does it help in standardizing cloud services and enabling interoperability? | 8 | L2 | 3 | 1,2 |
| b) | Explain the various service models and deployment models in cloud computing. | 8 | L2 | 3 | 1,2 |
| 5. a) | Explain the Google App Engine platform architecture with neat diagram. | 8 | L2 | 3 | 1,2 |
| b) | Compare Amazon Web Services, Google AppEngine, and Microsoft Azure in terms of services offered, scalability, and use cases. | 8 | L2 | 3 | 1,2 |
| 6. a) | Describe how cloud computing technology can be applied to support remote ECG monitoring. | 8 | L2 | 4 | 1,2 |
| b) | Explain the application of cloud computing technology to protein structure prediction. | 8 | L2 | 4 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Explain High Level Architecture framework for green cloud computing with neat diagram. | 8 | L2 | 5 | 1,2 |
| b) | Explain Market-oriented cloud computing scenario with a neat diagram. | 8 | L2 | 5 | 1,2 |
| 8. a) | Define federated clouds (inter-cloud). Explain their advantages in terms of scalability and resource sharing. Discuss the role and risks of third-party cloud services in enterprise adoption. | 8 | L2 | 5 | 1,2 |
| b) | Define microservice. Explain with a neat diagram the Monolithic architecture. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
